

RESEARCH FOCUS	Intersection of AI and statistics across Representation Learning, Interpretability, and LLMs	
CONTACT	taehyo97@gmail.com 917-815-6831	New York, NY
EDUCATION	New York University , New York, NY	
	Ph.D. in Biostatistics (Committee: Hai Shu, Yang Feng, Wen Zhou)	Expected Spring 2027
	M.S. in Computer Science	2022
	University of Toronto , Toronto, ON	
	B.A.Sc in Computer Engineering, Biomedical Engineering minor	2020
PROFESSIONAL EXPERIENCE	Machine Learning Research Intern	Summer 2026
	Dandy	New York, NY
	- Proposed a preference learning framework for 3D dental crown quality using VAE and Bradley Terry ranking head, improving the unguided manufacturing rate from 36% to 66%. - Represented 3D data in a topology preserving 2D latent space, accelerating training by 4x.	
	Statistical Machine Learning Researcher	Sept 2020 - present
	NYU School of Global Public Health	New York, NY
	Thesis title: Advancing statistical machine learning for Alzheimer’s disease research - Designed computationally efficient, spatial false discovery rate control methods for high dimensional neuroimaging, adapting unsupervised W-Net and hidden Markov random fields. - Formulated an L0 penalized sparse canonical correlation analysis method for interpretable multi-modal representation learning with application to imaging-omics data. - Proposed knowledge augmented, interpretable survival modeling framework utilizing LLMs.	
	NYU Steinhardt School of Culture, Education, and Human Development	
	Developed a self-supervised learning pipeline for lingual ultrasound video analysis.	
	NYU College of Dentistry	
	Conducted data curation and statistical analysis for clinical dental research.	
	NYU Courant Institute School of Mathematics, Computing, and Data Science	
	Built an Apple Watch system to detect, track, and help alleviate pediatric anxiety attacks.	
	FDA-OCE-ASA Oncology Educational Fellow	Oct 2025 - May 2026
	U.S. Food and Drug Administration (FDA)	Remote
	Evaluated oncology clinical trial evidence and regulatory decision-making through FDA-led case studies and a mock Oncologic Drugs Advisory Committee.	
	Deep Learning Researcher	Sept 2024 - May 2026
	Biofeedback Intervention Technology for Speech Lab, NYU Steinhardt	New York, NY
	- Presented the first self-supervised learning framework that focuses on within-rhotic articulatory structure in American English from lingual ultrasound video of children with residual SSD. - Integrated speaker-adversarial training that outperforms BYOL, VideoMAE, and RESA by 8% accuracy and 0.194 MCC on within-phoneme structure classification.	
	Statistical Fellow	Summer 2024
	Biostatistical Collaboration and Consultation Core, NYU GPH	New York, NY
	Consulted clients to define statistical analysis plans and supported manuscript development.	

Software Developer Intern

Summer 2019

N.1 Institute for Health, National University of Singapore

Kent Ridge, Singapore

- Assembled a parallel-distributed high-performance cluster system via HTCondor, Unix, and MATLAB to enable high throughput analysis of 76TB neural coding data.

Software Developer Intern

May 2018 - May 2019

Seiko Epson

Markham, ON

- Developed object tracking ground truth generation system by calibrating OptiTrack Motion Capture and Unity3D in real time.
- Implemented the Windows Software Development Kit (SDK) supporting Epson's augmented reality smart-glasses.

SKILLS

Programming: Python, PyTorch/JAX/Tensorflow, C/C++/C#, R, SQL, Java**MLOps:** MLflow, Flyte, Vertex AI, Spark, Docker, Kubernetes, Git, BigQuery, LangChain

PUBLICATIONS

Kim, T., Shu, H., Jia, Q., de Leon, M. J. DeepFDR: A Deep Learning-based False Discovery Rate Control Method for Neuroimaging Data. (2024).

AISTATS, 238: 946–954. [Runner-up Winner of JSM Student paper competition](#) [Paper] [Code]

Kim, T., Jia, Q., de Leon, M. J., Shu, H. A False Discovery Rate Control Method Using a Fully Connected Hidden Markov Random Field for Neuroimaging Data. (2026).

Medical Image Analysis, 104158. [Paper] [Code]

Kim, T., Eads, A., McFee, B., McAllister, T., Shu, H. Self-Supervised Learning on Lingual Ultrasound Video Encodes Clinically Meaningful Articulatory Structure of Rhotic Production in Residual Speech Sound Disorder. (2026).

MICCAI, In Press. [Paper] [Code]

Chen, Y., Kim, T., Shu, H., Feng Y. TabSODA: Tabular Diffusion Imputation Framework with Automatic Skip Detection and Ordinal Awareness. (2026).

Statistics and Trustworthy AI for Cross Domain Acceleration, In Press. [Paper] [Code]

Tang, T., Chen, Y., Kim, T., Shu, H. UKAN-EP: Enhancing U-KAN with Efficient Attention and Pyramid Aggregation for 3D Multi-Modal MRI Brain Tumor Segmentation. (2025).

BMC Medical Imaging, 25, 517. [Paper] [Code]

Hosseini, M. S., Bejnordi, B. E., Trinh, V. Q., Chan, L., Hasan, D., Li, X., Yang, S., Kim, T. et al. Computational Pathology: A Survey Review and the Way Forward. (2024).

Journal of Pathology Informatics, 15, 100357. [Paper]

SUBMITTED OR
UNDER REVISION

Eads, A., Kim, T., et al. Identifying subtypes of rhotic distortions using self-supervised learning on lingual ultrasound data. (2026).

Minor Revision at Journal of Speech, Language, and Hearing Research.

Kim, T., Shu, H. SKARF: LLM-based Knowledge Augmentation of Random Forests for Enhanced Survival Prediction. (2026).

Submitted to TLMR.

Li, B., Kim, T., Shu, H. DePHI: Dynamically Evolving System Framework for PHI-Safe and Auditable LLM Generation. (2026).

Submitted to Neurips 2026.

Lai, A., Kim, T., Dahlen, A., Lomas, T. A Global Understanding of Work Enjoyment and Human

Wellbeing. (2025).
Round 1 Review at Social Indicators Research.

Kim, H., Cardoso, D. d. M., Kayahara, G. M., Kim, T. et al. Phenotyping Neuropathic-Like Pain and Functional Impairment in Head and Neck Cancer Across Disease and Treatment States using a Novel Orofacial Sensory Phenotyping Questionnaire. (2026).
Submitted to British Journal of Anesthesia.

CURRENT PROJECTS	Kim, T., Shu, H. L0-IPLS: An L0 Penalized Sparse Canonical Correlation Analysis Method with Application to Imaging-Omics Data. <i>In preparation for Biometrics.</i>	
MEMBERSHIP AND AFFILIATIONS	Member, American Statistical Association	2022 - present
	Member, Eastern North American Region of the International Biometric Society	2022 - present
	Member, Korean International Statistical Society	2024 - present
	Reviewer, Medical Image Analysis	September 2026
	Reviewer, Computational Statistics & Data Analysis	August 2026
	Reviewer, IEEE International Conference on Data Mining (ICDM)	July 2026
	Reviewer, International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)	Mar 2026
	Reviewer, The American Statistician	Dec 2025
AWARDS	Reviewer, The American Statistician	Aug 2025
	FDA-OCE-ASA Oncology Educational Fellowship, FDA	2026
	JSM Student Paper Award, ASA Statistics in Imaging	2024
	DataFest Finalist, ENAR	2024
	Student Travel Award, American Statistical Association	2024
	PhD Fellowship Award, New York University	2022
	Certificate for Artificial Intelligence Engineering, University of Toronto	2020
	International Summer Research Award, University of Toronto	2019
CONFERENCE PRESENTATION	Gold Award, The Duke of Edinburgh's Award	2015
	"A False Discovery Rate Control Method Using a Fully Connected Hidden Markov Random Field for Neuroimaging Data"	
	Invited Oral, ASA SLDS, Brooklyn, NY	November 2026
	"Self-Supervised Learning on Lingual Ultrasound Video Encodes Clinically Meaningful Articulatory Structure of Rhotic Production in Residual Speech Sound Disorder"	
	Poster, MICCAI, Strasbourg, France	September 2026
	"A Global Understanding of Work Enjoyment and Human Wellbeing"	
	Poster, Joint Statistical Meetings, Nashville, TN	August 2025
	"Causal Determinants of Blood Pressure Control among US Adults with Hypertension: A Data Driven Causal Graphical Learning, NHANES 2013 to 2023"	
	Poster, ENAR Spring Meeting, New Orleans, LA	March 2025
	"DeepFDR: A Deep Learning-based False Discovery Rate Control Method for Neuroimaging Data"	
	Invited Oral, Joint Statistical Meetings, Portland, OR	August 2024
	"DeepFDR: A Deep Learning-based False Discovery Rate Control Method for Neuroimaging Data"	

Poster, International Conference in Artificial Intelligence and Statistics, Valencia, Spain May 2024

“Enhancing AI-based Speech Therapy through Acoustic to Articulatory Mapping”
 Poster, AI Research Symposium: Bridging AI Innovation and Societal Impact
 New York, NY April 2024

“Machine Learning-driven Risk Factor Identification on Post-2013 Blood Pressure Control Decline
 in Hypertensive Populations”
 Oral, ENAR Spring Meeting, Baltimore, MD March 2024

TEACHING
EXPERIENCE

Teaching Assistant, New York University, New York, NY

Statistical Methods in Genomics and Bioinformatics (GPH-GU 2378) Spring 2026
 Survey Design, Analysis, and Reporting (GPH-GU 2387) Fall 2025
 Applied Bayesian Analysis in Public Health (GPH-GU 2272/3372) Fall 2024
 Applied Survival Analysis (GPH-GU 2368/3368) Spring 2024
 Statistical Inference (GPH-GU 3225) Fall 2023

Graduate Student Mentor, New York University, New York, NY

Pathways into Quantitative Aging Research Summer Program Summer 2024
 Pathways into Quantitative Aging Research Summer Program Summer 2022